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CUET UG

Mathematics

Linear Programming

Formulation, Feasible Regions, Corner Point Theorem, Unbounded Half-Plane Tests & Optimization

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Syllabus & Exam Weightage

In the NTA CUET (UG) Mathematics examination, **Linear Programming** is a core component appearing in both **Section A** (the compulsory common section for all candidates) and **Section B1** (Core Mathematics). Typically accounting for **3 to 5 questions** (15 to 25 marks), this chapter offers unmatched scoring efficiency for students who master the geometric and algebraic principles of two-variable convex optimization.

Sub-Topic	Official NTA Syllabus Coverage	High-Yield CUET Focus
Mathematical Formulation	Terminology: Constraints, objective function, optimization, non-negativity; types of LPP: manufacturing, diet, transportation, investment.	Identifying correct inequalities, unit conversions (hours to minutes), "at least" ($\geq$) vs "at most" ($\leq$).
Graphical Method	Graphing linear inequalities in two variables, determining feasible and infeasible regions, identifying bounded vs unbounded regions.	Locating intersection coordinates of boundary lines, shading feasible polygon, non-negativity quadrant limits.
Corner Point Theorem	Fundamental Theorems of LPP (Theorems 1 & 2), evaluating objective function at vertices, finding maximum and minimum values.	Systematic tabular evaluation of $Z = ax + by$, identifying unique optimal vertex.
Unbounded Feasible Regions	Criteria for existence of optimal solutions in open/unbounded regions; the Open Half-Plane Test.	Testing $ax + by > M$ or $ax + by < m$; verifying if open half-plane intersects feasible region.
Special Scenarios	Multiple optimal solutions (infinite optima), unbounded solutions, infeasible problems (empty region), redundant constraints.	Matching slope of objective line with boundary constraint ($m_Z = m_L$), identifying redundant lines.

CUET EXAM STRATEGY — SPEED & PRECISION

CUET questions on Linear Programming are designed to test your understanding of **boundary slopes**, **vertex coordinates**, and **the open half-plane test** rather than tedious manual graphing. By understanding how the slope of the objective function $m = -\frac{a}{b}$ sweeps across the feasible polygon, you can often identify the optimal corner point in under 45 seconds without drawing the full graph.

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1. Mathematical Architecture of an LPP

Definition & The Four Pillars

A **Linear Programming Problem (LPP)** is an optimization problem concerned with finding the optimal (maximum or minimum) value of a linear function of several variables, subject to a collection of

linear equality or inequality constraints. In the CUET UG syllabus, all problems are strictly confined to **two decision variables** (x and y), permitting complete geometric analysis in the Cartesian plane $\mathbb{R}^2$.

Every well-posed LPP consists of four essential structural elements:

1. **Decision Variables (x, y):** The mathematical unknowns representing physical quantities to be produced, purchased, or allocated (e.g., number of tables and chairs, kilograms of wheat and rice).
2. **Objective Function (Z):** The linear criterion function to be optimized:
$$Z = ax + by$$
where $a, b \in \mathbb{R}$ are constants denoting marginal profit, cost, or yield per unit of x and y.
3. **Structural Constraints:** Linear inequalities representing physical, financial, or operational limitations:
$$a_i x + b_i y \leq c_i \quad \text{or} \quad a_i x + b_i y \geq c_i \quad (i = 1, 2, ..., m)$$
4. **Non-Negativity Restrictions:**
$$x \geq 0, \quad y \geq 0$$
Physical products cannot exist in negative quantities. Geometrically, these constraints restrict the search space entirely to the **First Quadrant** of the Cartesian plane.

THE LINEARITY AXIOM

In a linear programming model, two mathematical assumptions must strictly hold:

1. **Proportionality:** The contribution of each variable to both the objective function and the constraints is strictly proportional to its value. No terms like x^2, $\sqrt{x}$, or xy are permitted.
2. **Additivity:** The total contribution of all variables is the direct algebraic sum of their individual contributions. Synergistic or antagonistic non-linear interactions are excluded.

Standard Problem Archetypes in CUET

Linear programming problems in competitive examinations arise from four classical real-world operational scenarios:

Problem Archetype	Objective Nature	Typical Structural Constraints
Manufacturing Problem	Maximize Profit: $Z = p_1x + p_2y$	Machine hours available, raw material inventory limits, labor hour ceilings (all $\leq$).
Diet / Nutrition Problem	Minimize Cost: $Z = c_1x + c_2y$	Minimum required daily intake of vitamins, proteins, calories (all $\geq$).
Storage / Allocation Problem	Maximize Profit: $Z = p_1x + p_2y$	Total investment capital limit ($c_1x + c_2y \leq C$), storage capacity ($x + y \leq S$).
Transportation Problem	Minimize Shipping Cost	Factory supply capacities ($\leq$), warehouse destination demands ($\geq$).

CUET PHRASING TRAP — INEQUALITY DIRECTION

- **"At most", "not more than", "maximum capacity", "available limit"** $\Longrightarrow$ Use $\leq$.
- **"At least", "not less than", "minimum requirement", "exceeds"** $\Longrightarrow$ Use $\geq$.

- **"Unit Mismatch Trap":** If labor is given in **hours** (e.g., 8 hours) but manufacturing time per item is given in **minutes** (e.g., 15 minutes for item A, 20 minutes for item B), convert hours to minutes before writing the inequality:

$$15x + 20y \leq 8 \times 60 \implies 15x + 20y \leq 480.$$

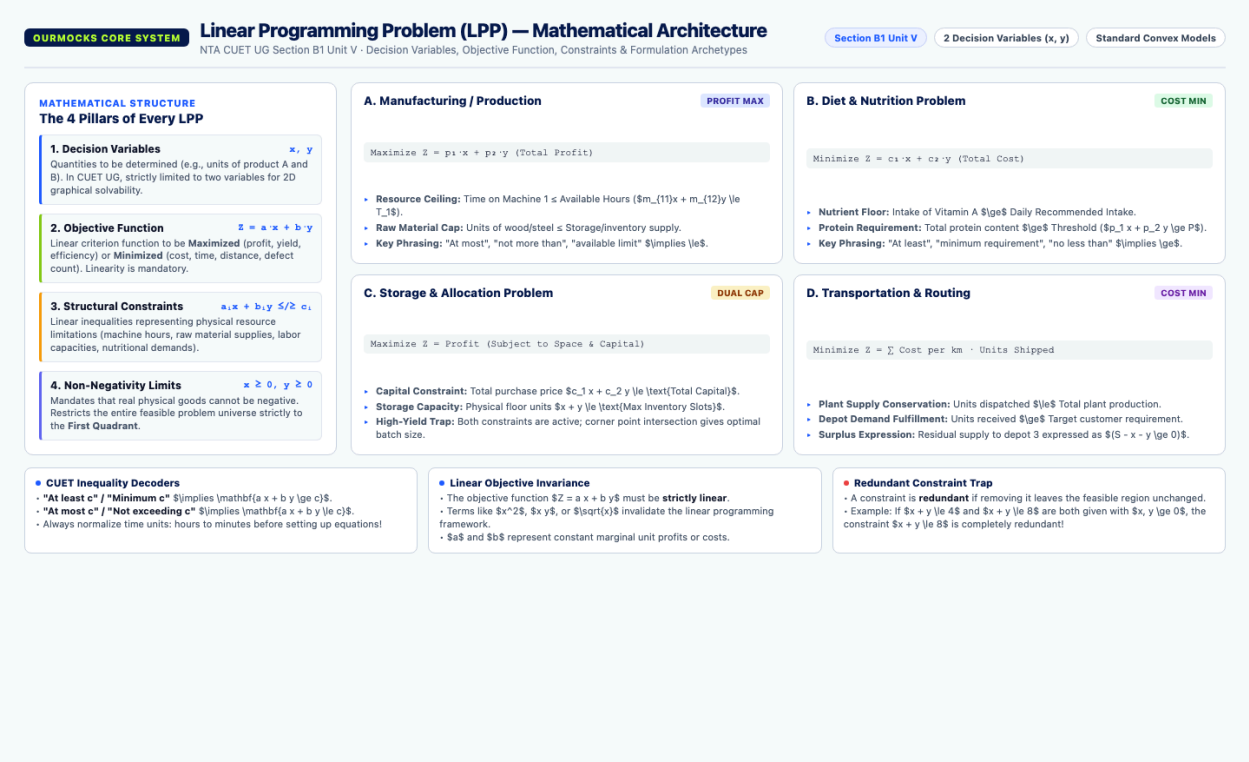


Figure 1: Figure 1: Mathematical Architecture of LPP, Core Pillars, Classical Archetypes, and Inequality Decoders

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2. Geometric Foundations: Convex Sets & Feasible Regions

Definitions & Terminology

- **Solution:** Any ordered pair (x, y) that satisfies the system of linear constraints is called a **solution**.
- **Feasible Region (R):** The common geometric region determined by the intersection of all constraints, including the non-negativity conditions $x \geq 0, y \geq 0$.
- **Feasible Solution:** Any point (x, y) that lies inside or on the boundary of the feasible region R. The feasible region contains infinitely many feasible points.
- **Infeasible Solution:** Any point lying outside the feasible region R. It violates at least one structural or non-negativity constraint.
- **Optimal Feasible Solution:** A feasible point $(x^*, y^*) \in R$ that produces the optimal (maximum or minimum) value of the objective function Z.

Convex Sets & The Polygonal Property

Definition: Convex Set Definition

A set of points $S \subseteq \mathbb{R}^2$ is said to be a **convex set** if for every pair of points $P_1, P_2 \in S$, the entire straight line segment joining P_1 and P_2 lies completely within S:

$$\lambda P_1 + (1-\lambda)P_2 \in S \quad \text{for all } \lambda \in [0,1].$$

Every closed half-plane defined by a linear inequality $ax + by \leq c$ is a convex set. Because the intersection of any collection of convex sets is itself convex, **the feasible region of every LPP is always a convex set**.

- If the feasible region is bounded by straight lines, it forms a **convex polygon**.
- In a convex polygon, all interior angles are strictly less than or equal to 180°. There are no indentations or re-entrant corners.

Bounded vs Unbounded Feasible Regions

1. **Bounded Feasible Region:** A feasible region that can be enclosed completely within a circle of finite radius r. It has a closed polygonal perimeter with a finite maximum diameter.
2. **Unbounded Feasible Region:** A feasible region that extends indefinitely into space in at least one direction. It cannot be enclosed within any finite circle.

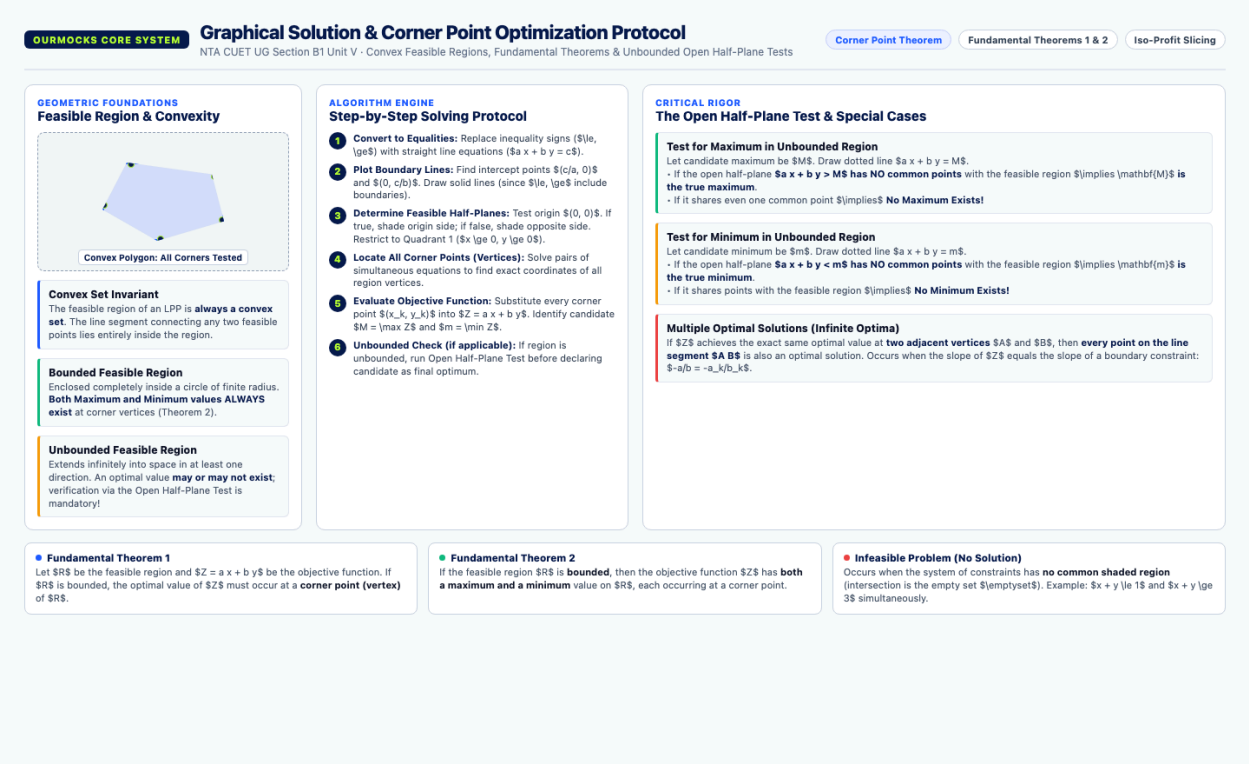


Figure 2: Figure 2: Convex Polygon Feasible Regions, Corner Point Protocol, and Unbounded Open Half-Plane Tests

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3. The Fundamental Theorems of Linear Programming

The entire mathematical validity of graphical optimization rests upon two foundational theorems established in convex analysis:

FUNDAMENTAL THEOREM 1 (THE CORNER POINT INVARIANT)

Let R be the feasible region (convex polygon) for a linear programming problem, and let $Z = ax + by$ be the objective function. If Z has an optimal value (maximum or minimum), where the variables are subject to linear constraints, this optimal value **must occur at an extreme point (corner point or vertex)** of the feasible region.

Mathematical Intuition: The level curves (iso-profit or iso-cost lines) of $Z = ax + by$ are parallel straight lines of constant slope $m = -\frac{a}{b}$. As the value of Z increases, the line translates across the plane. The last point of contact between the moving straight line and the convex polygon before leaving the region must be an **extreme vertex** (or an entire edge connecting two adjacent vertices).

FUNDAMENTAL THEOREM 2 (BOUNDED EXISTENCE GUARANTEE)

If the feasible region R for an LPP is **bounded**, then the objective function $Z = ax + by$ has **both a maximum and a minimum value** on R, and each of these values occurs at a corner point (vertex) of R.

CRUCIAL NUANCE FOR UNBOUNDED REGIONS

If the feasible region R is **unbounded**, an optimal value of Z **may or may not exist**.

- A maximum or minimum is NOT guaranteed.
- If an optimal value does exist, it must still occur at a corner point.
- However, you **cannot** merely take the largest or smallest value from the corner points table without verifying existence via the **Open Half-Plane Test**!

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4. The Corner Point Algorithm

For any two-variable LPP, the systematic corner point solving protocol proceeds through six precise mathematical stages:

Stage-by-Stage Protocol

1. **Stage 1 (Formulation):** State the decision variables x, y, write the objective function $Z = ax + by$, and translate all conditions into linear inequalities $a_i x + b_i y \leq c_i$ alongside $x \geq 0, y \geq 0$.
2. **Stage 2 (Boundary Lines):** Convert every inequality into a boundary equation $a_i x + b_i y = c_i$. Determine coordinate intercepts $\left(\frac{c_i}{a_i}, 0\right)$ and $\left(0, \frac{c_i}{b_i}\right)$ to draw the lines.
3. **Stage 3 (Half-Plane Selection):** Substitute the origin $(0, 0)$ into the inequality:
   - If $(0, 0)$ satisfies the inequality, shade the half-plane containing the origin.
   - If $(0, 0)$ does not satisfy it, shade the opposite half-plane.
   - **(Origin Line Exception):** If the boundary line passes through $(0, 0)$ (e.g., $y - 2x \leq 0$), test a distinct point such as $(1, 0)$ or $(0, 1)$ instead of the origin!
4. **Stage 4 (Locate Vertices):** Find the intersection coordinates of all boundary lines that define the vertices of the shaded feasible region R by solving simultaneous 2×2 linear systems.
5. **Stage 5 (Tabular Evaluation):** Compute the value of $Z = ax + by$ at each corner point (x_k, y_k):

Corner Point	Coordinates (x, y)	Value of $Z = ax + by$
A	(x_1, y_1)	$Z_1 = ax_1 + by_1$

Corner Point	Coordinates (x, y)	Value of $Z = ax + by$
B	(x_2, y_2)	$Z_2 = ax_2 + by_2$
C	(x_3, y_3)	$Z_3 = ax_3 + by_3$
D	(x_4, y_4)	$Z_4 = ax_4 + by_4$

6. **Stage 6 (Conclusion & Verification):**
   - If R is **bounded**, the largest value M is the maximum and the smallest value m is the minimum.
   - If R is **unbounded**, apply the Open Half-Plane Test to validate candidate M or m.

The Iso-Profit / Iso-Cost Lines Method (Sliding Line Technique)

While the Corner Point Method is universally applicable, an alternative powerful visual method is the **Iso-Profit (or Iso-Cost) Line Method**:

1. **Definition of Level Lines:** For any arbitrary constant k, the equation $ax + by = k$ defines a straight line known as an **iso-profit line** (if maximizing profit) or an **iso-cost line** (if minimizing cost).
2. **Constant Slope Invariant:** Every iso-profit line possesses the exact same slope:
$$m_Z = -\frac{a}{b}$$
Varying the value of k translates the straight line parallel to itself across the Cartesian plane without rotating it.
3. **The Sweeping Protocol:**
   - **For Maximization:** Draw an initial line $ax + by = k_0$ passing through the feasible region. Translate this line parallel to itself in the direction of increasing k (outward from the origin if $a, b > 0$). The **last corner point** of the feasible region touched by the line before it exits the region entirely is the **optimal maximizing vertex**.
   - **For Minimization:** Move the parallel line in the direction of decreasing k (inward towards the origin). The **first corner point** touched by the line as it enters the feasible region represents the **optimal minimizing vertex**.

CORNER POINT METHOD VS ISO-PROFIT LINE METHOD

- **Corner Point Method:** Computationally rigorous, algorithmic, ideal for formal examinations where coordinates of vertices must be tabulated and verified.
- **Iso-Profit Method:** Extremely fast for competitive multiple-choice exams like CUET when multiple choices represent corner points. Comparing the slope $m_Z = -\frac{a}{b}$ with the slopes of adjacent constraint boundaries allows you to instantly determine which vertex is extreme without calculating Z at every single point!

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5. The Open Half-Plane Test for Unbounded Regions

When the feasible region extends infinitely, an objective function designed for maximization may grow without bound ($Z \to \infty$), or a minimization objective may decrease without bound. To rigorously prove whether a candidate value from the corner points is the true global optimum, apply the **Open Half-Plane Test**:

Verification of Maximum Value (M)

1. Let M be the largest value among all corner points of the unbounded region.
2. Construct the open linear inequality:
$$ax + by > M$$
3. Graph the line $ax + by = M$ as a **dashed (open) line** across the Cartesian plane.
4. Determine the open half-plane corresponding to $ax + by > M$:
   - **Case 1 (True Maximum):** If the open half-plane $ax + by > M$ has **NO point in common** with the feasible region, then $\boldsymbol{M}$ **is the true, valid maximum value** of Z.
   - **Case 2 (No Maximum):** If the open half-plane $ax + by > M$ shares **even a single point** with the feasible region, then Z can attain values strictly greater than M. Hence, $\boldsymbol{Z}$ **has no maximum value** (the solution is unbounded).

Verification of Minimum Value (m)

1. Let m be the smallest value among all corner points of the unbounded region.
2. Construct the open linear inequality:
$$ax + by < m$$
3. Graph the line $ax + by = m$ as a **dashed (open) line**.
4. Determine the open half-plane $ax + by < m$:
   - **Case 1 (True Minimum):** If the open half-plane $ax + by < m$ has **NO point in common** with the feasible region, then $\boldsymbol{m}$ **is the true, valid minimum value** of Z.
   - **Case 2 (No Minimum):** If the open half-plane $ax + by < m$ shares any point with the feasible region, then $\boldsymbol{Z}$ **has no minimum value**.

UNBOUNDED MINIMIZATION WITH OPEN HALF-PLANE TEST

Problem: Minimize $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, and $x, y \geq 0$.

Step 1 (Vertices):

- Line 1: $x + 3y = 3$ intersects axes at $(3, 0)$ and $(0, 1)$.
- Line 2: $x + y = 2$ intersects axes at $(2, 0)$ and $(0, 2)$.
- Intersection of lines: Subtracting equations gives $2y = 1 \implies y = \frac{1}{2}, x = \frac{3}{2}$.
- Feasible region is unbounded (lies above both lines in Quadrant 1).
- Corner points are $A(0, 2)$, $B\left(\frac{3}{2}, \frac{1}{2}\right)$, and $C(3, 0)$.

Step 2 (Evaluate Z at Corners):

- At $A(0, 2)$: $Z = 3(0) + 5(2) = 10$.
- At $B\left(\frac{3}{2}, \frac{1}{2}\right)$: $Z = 3\left(\frac{3}{2}\right) + 5\left(\frac{1}{2}\right) = \frac{9}{2} + \frac{5}{2} = \frac{14}{2} = 7$.
- At $C(3, 0)$: $Z = 3(3) + 5(0) = 9$.
- Candidate minimum is $m = 7$ at $B\left(\frac{3}{2}, \frac{1}{2}\right)$.

Step 3 (Open Half-Plane Test):

- Formulate inequality: $3x + 5y < 7$.
- Test origin $(0, 0)$: $3(0) + 5(0) = 0 < 7$ (True). The half-plane lies towards the origin.
- The line $3x + 5y = 7$ passes through $\left(\frac{7}{3}, 0\right) \approx (2.33, 0)$ and $\left(0, \frac{7}{5}\right) = (0, 1.4)$.
- It touches the feasible region only at the single vertex $B\left(\frac{3}{2}, \frac{1}{2}\right)$. But the open half-plane $3x + 5y < 7$ **excludes the boundary line itself!**
- Therefore, the open half-plane has **no points in common** with the feasible region.

Conclusion: The minimum value of Z exists and is strictly **7** at $\left(\frac{3}{2}, \frac{1}{2}\right)$.

UNBOUNDED MAXIMIZATION FAILURE CASE (NO MAXIMUM EXISTS)

Problem: Maximize $Z = x + y$ subject to $x - y \geq 0$, $-x + 2y \geq 0$, and $x, y \geq 0$.

Step 1 (Feasible Geometry):

- Line 1: $y = x$ (bisector of Quadrant 1). The condition $x - y \geq 0 \implies y \leq x$ shades the region below the line $y = x$.
- Line 2: $y = \left(\frac{1}{2}\right)x$. The condition $-x + 2y \geq 0 \implies 2y \geq x \implies y \geq \left(\frac{1}{2}\right)x$ shades the region above the line $y = \left(\frac{1}{2}\right)x$.
- Both conditions together with $x, y \geq 0$ define an infinite open wedge between the rays $y = \left(\frac{1}{2}\right)x$ and $y = x$ extending infinitely away from the origin into the first quadrant.
- The only corner point of this region is the origin $O(0, 0)$.

Step 2 (Corner Evaluation):

- At $O(0, 0)$, $Z = 0 + 0 = 0$. Candidate maximum is $M = 0$.

Step 3 (Open Half-Plane Test):

- Construct open inequality $x + y > 0$.
- Every point in the interior of the feasible wedge (for instance, $(4, 3)$, which satisfies $3 \leq 4$ and $2(3) = 6 \geq 4$) gives $Z = 4 + 3 = 7 > 0$.
- Clearly, the open half-plane $x + y > 0$ contains infinitely many points of the feasible region!

Conclusion: The objective function Z can grow arbitrarily large as (x, y) moves outward along the wedge. Therefore, ***Z* has NO maximum value** (Unbounded solution).

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6. Special Phenomena & Geometric Anomalies

CUET examination papers frequently test four anomalous situations where standard single-point optimization behaves unusually:

1. Multiple Optimal Solutions (Infinite Optima)

If the objective function $Z = ax + by$ achieves the exact same optimal value at **two distinct adjacent corner points** $A(x_1, y_1)$ and $B(x_2, y_2)$ of the feasible region:

$$Z(A) = Z(B) = Z^*$$

Then **every single point on the line segment *AB* connecting them is also an optimal solution**, giving rise to an **infinite number of optimal solutions**.

GEOMETRIC CONDITION FOR INFINITE OPTIMA

Multiple optimal solutions occur if and only if:

1. The objective function is parallel to the boundary constraint line connecting vertices A and B.
2. The slope of the objective function equals the slope of the binding constraint:

$$m_Z = -\frac{a}{b} = -\frac{a_k}{b_k} = m_k$$

2. Infeasible Problem (No Feasible Solution)

When there is no single point that simultaneously satisfies all given constraints along with $x \geq 0, y \geq 0$, the feasible region is the **empty set** ($R = \varnothing$):

- Geometrically, the half-planes point in contradictory directions and share zero overlapping area.
- **Conclusion:** The problem has **no feasible solution**.

3. Unbounded Objective Function

Occurs when the feasible region is unbounded and the objective function can grow indefinitely in the direction of optimization ($Z \to +\infty$ for maximization or $Z \to -\infty$ for minimization).

- In such cases, the open half-plane test fails, and the problem has **no finite optimal solution**.

4. Redundant Constraints

A constraint is termed **redundant** if its complete removal from the system of inequalities produces absolutely no change in the feasible region.

- Example: Subject to $x + y \leq 4$, $x + y \leq 10$, and $x, y \geq 0$.
- Here, any point satisfying $x + y \leq 4$ automatically satisfies $x + y \leq 10$. The constraint $x + y \leq 10$ is completely inactive and redundant.

5. Reduction of Transportation Problems to Two Variables

In operations research, transportation problems typically involve m factories and n depots, creating $m \times n$ variables. In CUET UG, these are systematically reduced to **two decision variables** (x and y) using the **principle of conservation of supply and demand**:

- Let Factory 1 have capacity S_1 and Factory 2 have capacity S_2.
- Let Depot A require D_1, Depot B require D_2, and Depot C require D_3.
- Let x units be shipped from Factory 1 to Depot A, and y units from Factory 1 to Depot B.
- Then the remaining supply shipped from Factory 1 to Depot C is automatically $(S_1 - x - y)$.
- Depot A needs D_1, so Factory 2 must supply $(D_1 - x)$.
- Depot B needs D_2, so Factory 2 must supply $(D_2 - y)$.
- Depot C needs D_3, so Factory 2 supplies $S_2 - (D_1 - x) - (D_2 - y)$.
- **Constraints generated:** All shipment quantities must be non-negative:

$$x \geq 0, \quad y \geq 0, \quad x + y \leq S_1, \quad x \leq D_1, \quad y \leq D_2$$

This converts an apparently complex logistics network into a standard two-variable convex polygon!

6. Sensitivity Analysis & Objective Slope Tolerances

A common higher-order thinking skill tested in CUET is determining the **range of values** that an objective coefficient a or b can take without shifting the optimal vertex:

- Suppose the optimal vertex V is formed by the intersection of two boundary constraint lines L_1 (slope m_1) and L_2 (slope m_2), with $m_1 < m_2$.
- The slope of the objective function is $m_Z = -\frac{a}{b}$.
- The vertex V remains the unique optimal solution as long as the slope of Z stays strictly bounded between the slopes of the two intersecting constraint edges:

$$m_1 \leq -\frac{a}{b} \leq m_2$$

- If $-\frac{a}{b}$ equals m_1, the solution degenerates into infinite optimal solutions along boundary L_1. If $-\frac{a}{b} < m_1$, the optimal solution jumps to an adjacent corner point!

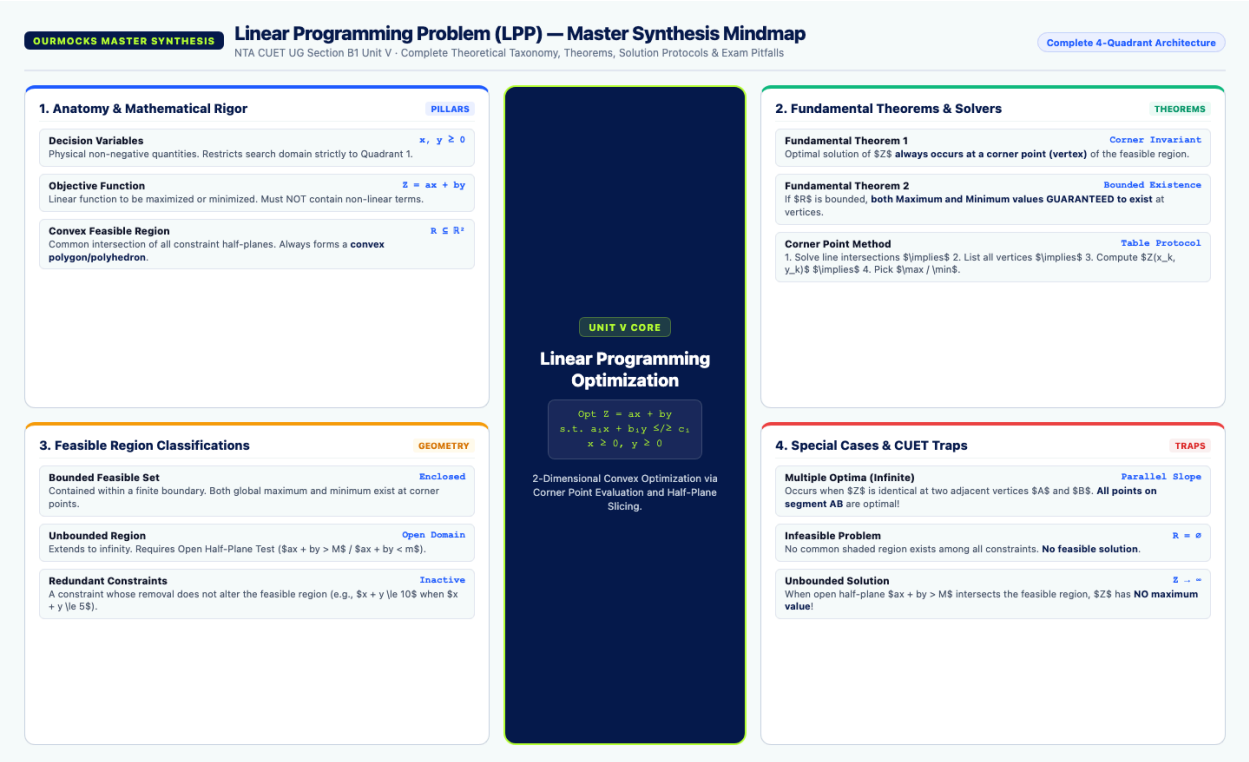


Figure 3: Figure 3: Master Synthesis Mindmap of Linear Programming Taxonomy, Theorems, and Traps

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7. High-Yield Formula Cheatsheet & Diagnostic Traps

MASTER FORMULA & CONCEPT CHEATSHEET — LINEAR PROGRAMMING

Core Equations & Invariants:

- Objective Function: $Z = ax + by$ (strictly linear; slope $m = -\frac{a}{b}$).
- Convex Feasible Set: Line segment connecting any two points in R lies entirely in R.
- Corner Point Theorem: Optimal solution occurs at an extreme vertex of R.
- Bounded Region: Both Maximum and Minimum guaranteed to exist.
- Infinite Optima: $Z(A) = Z(B) \implies$ All points on segment AB yield Z^*.

Unbounded Region Protocol:

- Maximum M: Check $ax + by > M$. If no common point with $R \implies M$ is maximum.
- Minimum m: Check $ax + by < m$. If no common point with $R \implies m$ is minimum.

Key Phrasing Translation Table:

- "At least k" $\implies ax + by \geq k$
- "At most k" $\implies ax + by \leq k$
- "Exceeds by at least d" $\implies x - y \geq d$
- "Ratio of x to y is at least r" $\implies \frac{x}{y} \geq r \implies x - ry \geq 0$

FIVE FATAL CUET EXAM TRAPS IN LPP

1. **The Origin Boundary Trap:** If a constraint is $y \leq 2x$, the boundary line $y = 2x$ passes through $(0, 0)$. Testing $(0, 0)$ gives $0 \leq 0$ (uninformative identity). You must test $(1, 0)$ (gives $0 \leq 2$, true $\implies$ shade towards positive x-axis).

2. **The Non-Negativity Blindspot:** Candidates frequently forget that $x \geq 0, y \geq 0$ chops off all quadrants except the First Quadrant! Never include negative coordinate vertices.
3. **The Parallel Slope Shortcut:** If $Z = 4x + 6y$ and one of the boundary lines is $2x + 3y = 12$, notice that $Z = 2(2x + 3y) = 2(12) = 24$ along the entire edge! The problem automatically has **infinite optimal solutions**.
4. **Declaring Unbounded Maximum Without Testing:** In an unbounded region, finding candidate $M = 150$ at a corner point does NOT mean 150 is the maximum. Almost always, an unbounded maximization problem has **NO maximum** ($Z \to \infty$).
5. **Integer Rounding Trap:** If an LPP asks for optimal items, and the vertex is $(3.4, 2.8)$, do NOT merely round to $(3, 3)$ without checking whether $(3, 3)$ lies inside the feasible region!

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Practice Questions — Part 1 (Q1 to Q5)

Q1. Which of the following statements is universally TRUE for any linear programming problem?

(A) The feasible region is always a non-convex set.
(B) The objective function can be a non-linear quadratic function.
(C) The feasible region is always a convex polygon or convex polyhedron.
(D) An optimal solution can only occur in the interior of the feasible region.

Sol. Q1. The intersection of any finite number of closed half-planes defined by linear inequalities is always a convex set. Therefore, the feasible region of any LPP is guaranteed to be a convex polygon (if bounded) or convex polyhedron. By the Corner Point Theorem, optimal solutions always occur at extreme boundary vertices, never exclusively in the interior. **Answer: (C)**

Q2. In a linear programming problem, the objective function $Z = 3x + 2y$ is to be maximized subject to $x + y \leq 4$, $x - y \leq 2$, and $x \geq 0, y \geq 0$. The maximum value of Z is:

(A) 12
(B) 11
(C) 8
(D) 14

Sol. Q2. Find all corner points of the bounded feasible region:

1. Origin: $O(0,0) \Longrightarrow Z = 0$.
2. y-intercept of $x + y = 4$: $A(0,4) \Longrightarrow Z = 3(0) + 2(4) = 8$.
3. Intersection of $x + y = 4$ and $x - y = 2$: Adding gives $2x = 6 \Longrightarrow x = 3, y = 1$. At $B(3,1) \Longrightarrow Z = 3(3) + 2(1) = 9 + 2 = 11$.
4. x-intercept of $x - y = 2$: $C(2,0) \Longrightarrow Z = 3(2) + 2(0) = 6$.

The maximum value among all corner points is 11 at $(3,1)$. **Answer: (B)**

Q3. If the objective function $Z = ax + by$ achieves its maximum value at two distinct corner points $A(2,3)$ and $B(4,1)$ of a bounded feasible region, then the relationship between a and b is:

(A) $a = b$
(B) $a = 2b$
(C) $2a = b$
(D) $a + b = 0$

Sol. Q3. Since Z achieves the same maximum value at both points:

$$Z(A) = Z(B) \Longrightarrow 2a + 3b = 4a + b$$

Rearranging terms:

$$3b - b = 4a - 2a \Longrightarrow 2b = 2a \Longrightarrow a = b.$$

Furthermore, every point on the line segment joining $(2,3)$ and $(4,1)$ is also an optimal solution, yielding an infinite number of optimal solutions. **Answer: (A)**

Q4. The corner points of the feasible region determined by a system of linear constraints are $(0,3)$, $(1,1)$, and $(3,0)$. Let $Z = px + qy$, where $p, q > 0$. The condition on p and q such that the minimum of Z occurs at both $(0,3)$ and $(1,1)$ is:

(A) $p = 2q$
(B) $q = 2p$
(C) $p = q$
(D) $p = 3q$

Sol. Q4. Equate the values of Z at the two given points:

$$Z(0,3) = p(0) + q(3) = 3q$$
$$Z(1,1) = p(1) + q(1) = p + q$$

Setting $Z(0,3) = Z(1,1)$:

$$3q = p + q \Longrightarrow 2q = p \Longrightarrow p = 2q.$$

This condition ensures that the slope of Z matches the slope of the line segment connecting $(0,3)$ and $(1,1)$. **Answer: (A)**

Q5. Consider the feasible region for an LPP shown to be unbounded with corner points $(0,4)$, $(2,2)$, and $(6,0)$. The objective function is $Z = 4x + 2y$. The candidate minimum is 12 at $(2,2)$. To confirm whether 12 is the true minimum, we must verify that:

(A) The open half-plane $4x + 2y > 12$ has points in common with the feasible region.
(B) The open half-plane $4x + 2y < 12$ has NO points in common with the feasible region.
(C) The line $4x + 2y = 12$ does not intersect the coordinate axes.
(D) The feasible region can be enclosed in a finite circle.

Sol. Q5. According to the Open Half-Plane Test for unbounded minimization problems, a candidate minimum m is the true minimum if and only if the strictly open half-plane $ax + by < m$ (here $4x + 2y < 12$) has **no points in common** with the feasible region. If it shares any common points, no minimum exists. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. The maximum value of $Z = 2x + 3y$ subject to constraints $x + 2y \leq 10$, $2x + y \leq 14$, and $x \geq 0, y \geq 0$ is:

(A) 16 **(B)** 18
(C) 19 **(D)** 20

Sol. Q6. Locate all corner points of the bounded region:

1. Origin: $O(0,0) \Longrightarrow Z = 0$.
2. y-intercept of $x + 2y = 10$: $(0,5) \Longrightarrow Z = 2(0) + 3(5) = 15$.
3. Intersection of $x + 2y = 10$ and $2x + y = 14$: Multiply first equation by 2: $2x + 4y = 20$. Subtract second equation: $3y = 6 \Longrightarrow y = 2, x = 10 - 2(2) = 6$. At $(6,2) \Longrightarrow Z = 2(6) + 3(2) = 12 + 6 = 18$.
4. x-intercept of $2x + y = 14$: $(7,0) \Longrightarrow Z = 2(7) + 3(0) = 14$.

The maximum value is 18 occurring at $(6,2)$. **Answer: (B)**

Q7. A constraint in an LPP is given as: "The quantity of product A produced must be at least twice the quantity of product B produced." The correct algebraic inequality representing this constraint is:

(A) $x - 2y \leq 0$ **(B)** $x - 2y \geq 0$
(C) $2x - y \geq 0$ **(D)** $x + 2y \geq 0$

Sol. Q7. Let x be the quantity of product A and y be the quantity of product B. "At least twice" means $x \geq 2y$. Rearranging this inequality by subtracting $2y$ from both sides yields:

$$x - 2y \geq 0.$$

(Option A reverses the inequality, while Option C incorrectly doubles product A). **Answer: (B)**

Q8. The system of linear inequalities $x + y \leq 1$, $x + y \geq 3$, and $x \geq 0, y \geq 0$ represents a linear programming problem with:

(A) A unique optimal solution **(B)** Infinitely many optimal solutions
(C) An unbounded optimal solution **(D)** No feasible solution

Sol. Q8. The inequality $x + y \leq 1$ represents the region on and below the line $x + y = 1$ (towards the origin). The inequality $x + y \geq 3$ represents the region on and above the line $x + y = 3$ (away from the origin). These two parallel half-planes do not intersect anywhere. Since the feasible region is empty ($R = \varnothing$), the problem has **no feasible solution**. **Answer: (D)**

Q9. The corner points of a feasible region are $O(0,0)$, $A(5,0)$, $B(4,3)$, and $C(0,5)$. If the objective function is $Z = 200x + 500y$, the difference between the maximum and minimum values of Z is:

(A) 2300 **(B)** 2500
(C) 2000 **(D)** 1500

Sol. Q9. Evaluate $Z = 200x + 500y$ at each corner point:

- At $O(0,0)$: $Z = 0$.
- At $A(5,0)$: $Z = 200(5) + 500(0) = 1000$.
- At $B(4,3)$: $Z = 200(4) + 500(3) = 800 + 1500 = 2300$.
- At $C(0,5)$: $Z = 200(0) + 500(5) = 2500$.

Maximum value $M = 2500$ (at C), and Minimum value $m = 0$ (at O). The difference is $M - m = 2500 - 0 = 2500$. **Answer: (B)**

Q10. If an LPP has an unbounded feasible region and the candidate maximum value of $Z = ax + by$ obtained from corner points is M, which condition proves that Z has NO maximum value?

(A) The open half-plane $ax + by > M$ has at least one point in common with the feasible region. **(B)** The open half-plane $ax + by > M$ is disjoint from the feasible region.
(C) The line $ax + by = M$ passes through the origin. **(D)** The value of M is negative.

Sol. Q10. If the open half-plane $ax + by > M$ shares even a single point with the feasible region, it implies there exist feasible points where the objective function evaluates to values strictly greater than M. Since the region extends infinitely, Z can grow without bound. Hence, Z has **no maximum value**. **Answer: (A)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. In an LPP, the feasible region has corner points $(0,2)$, $(3,0)$, $(6,0)$, and $(0,5)$. If the objective function is $Z = 4x + 6y$, the minimum value of Z occurs at:

(A) Only $(0,2)$ **(B)** Only $(3,0)$
(C) Every point on the segment joining $(0,2)$ and $(3,0)$ **(D)** Only $(6,0)$

Sol. Q11. Evaluate $Z = 4x + 6y$ at the corner points:

- At $(0,2)$: $Z = 4(0) + 6(2) = 12$.
- At $(3,0)$: $Z = 4(3) + 6(0) = 12$.
- At $(6,0)$: $Z = 4(6) + 6(0) = 24$.
- At $(0,5)$: $Z = 4(0) + 6(5) = 30$.

The minimum value 12 is attained at two distinct adjacent corner points $(0,2)$ and $(3,0)$. By Theorem 1, Z attains this minimum value of 12 at **every point on the line segment** joining $(0,2)$ and $(3,0)$. **Answer: (C)**

Q12. A manufacturing firm produces two types of gadgets, X and Y. Each unit of X requires 2 hours on machine A and 1 hour on machine B. Each unit of Y requires 1 hour on machine A and 3 hours on machine B. Machine A can run for at most 10 hours/day and machine B for at most 12 hours/day. The constraints are:

(A) $2x + y \le 10,\quad x + 3y \le 12,\quad x, y \ge 0$ **(B)** $x + 2y \le 10,\quad 3x + y \le 12,\quad x, y \ge 0$
(C) $2x + y \ge 10,\quad x + 3y \ge 12,\quad x, y \ge 0$ **(D)** $2x + 3y \le 22,\quad x, y \ge 0$

Sol. Q12. Translate resource requirements for each machine:

- Machine A: 2 hrs per unit of X and 1 hr per unit of Y. Total time $2x + y$. Capacity is "at most 10 hours", so $2x + y \le 10$.
- Machine B: 1 hr per unit of X and 3 hrs per unit of Y. Total time $x + 3y$. Capacity is "at most 12 hours", so $x + 3y \le 12$.
- Non-negativity: $x \ge 0, y \ge 0$.

Matches Option A exactly. **Answer: (A)**

Q13. Which of the following geometric figures CANNOT be the feasible region of any linear programming problem?

(A) A bounded convex pentagon **(B)** An unbounded convex region in the first quadrant
(C) A non-convex polygon with an interior angle of 240° **(D)** A single line segment

Sol. Q13. The feasible region of an LPP is formed by the intersection of a finite number of linear half-planes. In Euclidean geometry, the intersection of convex sets is strictly convex. A non-convex polygon contains at least one interior angle $> 180^\circ$ (such as 240°), meaning a line segment connecting two internal points would pass outside the set. Hence, it can **never** be a feasible region. **Answer: (C)**

Q14. Given the objective function $Z = 5x + 3y$ subject to $x + y \le 60$, $x \ge 20$, $y \le 40$, and $x, y \ge 0$. The maximum value of Z is:

(A) 240 **(B)** 260
(C) 300 **(D)** 280

Sol. Q14. Determine all corner points of the feasible region:

1. $x = 20, y = 0 \Longrightarrow Z = 5(20) + 3(0) = 100$.
2. $x = 20, y = 40 \Longrightarrow Z = 5(20) + 3(40) = 100 + 120 = 220$.
3. Intersection of $y = 40$ and $x + y = 60 \Longrightarrow x = 20, y = 40$ (already counted).
4. $y = 40$ with $x + y \le 60$: For $y = 40$, maximum $x = 20$.
5. When $y = 0$, $x + y = 60 \Longrightarrow x = 60, y = 0$. Here $x = 60 \ge 20$ (valid). At $(60, 0) \Longrightarrow Z = 5(60) + 3(0) = 300$.
6. Check point where $x + y = 60$ and $y \le 40$: For instance, if $y = 40$, $x = 20$. If $y = 10$, $x = 50 \Longrightarrow Z = 5(50) + 3(10) = 280 < 300$.

The maximum value is 300 attained at $(60, 0)$. **Answer: (C)**

Q15. If a constraint in an LPP is changed from $x + y \le 10$ to $x + y \le 15$, while all other constraints and the objective function remain unchanged, the optimal value of a maximization objective function will:

(A) Always decrease **(B)** Either increase or remain unchanged
(C) Always remain strictly unchanged **(D)** Become zero

Sol. Q15. Relaxing a constraint (changing from ≤ 10 to ≤ 15) expands or leaves unchanged the existing feasible region R. Any previously feasible point remains feasible, and new points are added ($R_{\text{old}} \subseteq R_{\text{new}}$). Therefore, the maximum of the objective function over the expanded set can never be smaller than the maximum over the original subset. The optimal value will **either increase or remain unchanged** (if the original constraint was non-binding or redundant). **Answer: (B)**